CSC 370 PaperTrade, Functional Dependencies and Candidate Keys

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Overview

This document gives the functional dependencies (FDs) of each relation in the `stock_trading` database, and uses attribute closures to find every candidate key and count the superkeys. Relations and attributes match `sql/ddl.sql` and `docs/Schemata Overview/schemata.md`.

For a relation with attribute set C:

- X is a **superkey** if $X^+ = C$.
- X is a **candidate key** if it is a superkey and no proper subset of it is.
- **No-fire lemma:** if X contains no FD's left-hand side, then $X^+ = X$. In particular $\emptyset^+ = \emptyset$.

Glossary

Functional Dependencies

FD	Relation	Dependency
FD_1	`user`	user_id → name, email, cash_balance
FD_2	`user`	email → user_id
FD_3	`portfolio`	portfolio_id → user_id, portfolio_name, created_at
FD_4	`portfolio`	user_id, portfolio_name → portfolio_id
FD_5	`exchange`	exchange_code → exchange_name, country, currency
FD_6	`stock`	ticker → company_name, sector, exchange_code
FD_7	`stock_price_history`	ticker, price_date → opening_price, closing_price, high_price, low_price, volume
FD_8	`watchlist`	user_id, ticker → target_price, added_at
FD_9	`transaction`	transaction_id → portfolio_id, ticker, transaction_type, quantity, price, executed_at
FD_{10}	`limit_order`	order_id → portfolio_id, ticker, order_type, target_price, quantity, status, created_at

Keys

Relation	FDs	**Candidate keys** (primary key first)	**Superkeys**
`user`	FD_1, FD_2	{user_id}, {email}	12
`portfolio`	FD_3, FD_4	{portfolio_id}, {user_id, portfolio_name}	10
`exchange`	FD_5	{exchange_code}	8
`stock`	FD_6	{ticker}	8
`stock_price_history`	FD_7	{ticker, price_date}	32
`watchlist`	FD_8	{user_id, ticker}	4
`transaction`	FD_9	{transaction_id}	64
`limit_order`	FD_{10}	{order_id}	128

Every FD's left-hand side is a superkey of its relation, so every relation is in BCNF.

1 user

$$C_U = \{\text{user_id}, \text{name}, \text{email}, \text{cash_balance}\}$$

- FD_1: user_id $\to$ name, email, cash_balance. Each user ID is one account.
- FD_2: email $\to$ user_id. An email registers one account (C.1.2).

Closures. $\{\text{user_id}\}^+ = C_U$ by FD_1. $\{\text{email}\}^+ = C_U$ by FD_2 then FD_1.
Minimality. Both keys are single attributes and $\emptyset^+ = \emptyset$.
No others. A set with neither user_id nor email fires no FD, so every superkey contains one of them.
Candidate keys: {user_id} (primary), {email}. **Superkeys:** $2^3 + 2^3 - 2^2 = 12$.

2 portfolio

$$C_P = \{\text{portfolio_id}, \text{user_id}, \text{portfolio_name}, \text{created_at}\}$$

- FD_3: portfolio_id $\to$ user_id, portfolio_name, created_at. Each portfolio ID is one account.
- FD_4: user_id, portfolio_name $\to$ portfolio_id. A user's portfolio names are unique (C.2.4).

Closures. $\{\text{portfolio_id}\}^+ = C_P$ by FD_3. $\{\text{user_id}, \text{portfolio_name}\}^+ = C_P$ by FD_4 then FD_3.
Minimality. $\{\text{user_id}\}^+ = \{\text{user_id}\}$ and $\{\text{portfolio_name}\}^+ = \{\text{portfolio_name}\}$ by the no-fire lemma, so neither half of the composite key is a superkey.
No others. A set without portfolio_id and missing user_id or portfolio_name fires no FD.
Candidate keys: {portfolio_id} (primary), {user_id, portfolio_name}. **Superkeys:** $2^3 + 2^2 - 2^1 = 10$.

3 exchange

$$C_E = \{\text{exchange_code}, \text{exchange_name}, \text{country}, \text{currency}\}$$

- FD_5: exchange_code $\to$ exchange_name, country, currency. Each code is one exchange.

Closure. $\{\text{exchange_code}\}^+ = C_E$ by FD_5. Single attribute, so minimal. Any set without it fires no FD.
Candidate key: {exchange_code}. **Superkeys:** $2^3 = 8$.

4 stock

$$C_S = \{\text{ticker}, \text{company_name}, \text{sector}, \text{exchange_code}\}$$

- FD_6: ticker $\to$ company_name, sector, exchange_code. A ticker is one listed security on one exchange.

Closure. $\{\text{ticker}\}^+ = C_S$ by FD_6. Single attribute, so minimal. Any set without it fires no FD.
Candidate key: {ticker}. **Superkeys:** $2^3 = 8$.

5 stock_price_history

$$C_H = \{\text{ticker}, \text{price_date}, \text{opening_price}, \text{closing_price}, \text{high_price}, \text{low_price}, \text{volume}\}$$

- FD_7: ticker, price_date $\to$ opening_price, closing_price, high_price, low_price, volume. One price record per stock per day (C.5.1).

Closure. $\{\text{ticker}, \text{price_date}\}^+ = C_H$ by FD_7.
Minimality. $\{\text{ticker}\}^+ = \{\text{ticker}\}$ and $\{\text{price_date}\}^+ = \{\text{price_date}\}$ by the no-fire lemma.
No others. A set missing either attribute fires no FD.
Candidate key: {ticker, price_date}. **Superkeys:** $2^5 = 32$.

6 watchlist

$$C_W = \{\text{user_id}, \text{ticker}, \text{target_price}, \text{added_at}\}$$

- FD_8: user_id, ticker $\to$ target_price, added_at. A user watches a stock at most once (C.6.1).

Closure. $\{\text{user_id}, \text{ticker}\}^+ = C_W$ by FD_8.
Minimality. $\{\text{user_id}\}^+ = \{\text{user_id}\}$ and $\{\text{ticker}\}^+ = \{\text{ticker}\}$ by the no-fire lemma. FD_1 belongs to `user`, and its right-hand side is not in C_W.
No others. A set missing either attribute fires no FD.
Candidate key: {user_id, ticker}. **Superkeys:** $2^2 = 4$.

7 transaction

$$C_T = \{\text{transaction_id}, \text{portfolio_id}, \text{ticker}, \text{transaction_type}, \text{quantity}, \text{price}, \text{executed_at}\}$$

- FD_9: transaction_id $\to$ portfolio_id, ticker, transaction_type, quantity, price, executed_at. Each ID records one trade.

Closure. $\{\text{transaction_id}\}^+ = C_T$ by FD_9. Single attribute, so minimal. Any set without it fires no FD.
Candidate key: {transaction_id}. **Superkeys:** $2^6 = 64$.

8 limit_order

$$C_L = \{\text{order_id}, \text{portfolio_id}, \text{ticker}, \text{order_type}, \text{target_price}, \text{quantity}, \text{status}, \text{created_at}\}$$

- FD_{10}: order_id $\to$ portfolio_id, ticker, order_type, target_price, quantity, status, created_at. Each ID is one order.

Closure. $\{\text{order_id}\}^+ = C_L$ by FD_{10}. Single attribute, so minimal. Any set without it fires no FD.
Candidate key: {order_id}. **Superkeys:** $2^7 = 128$.